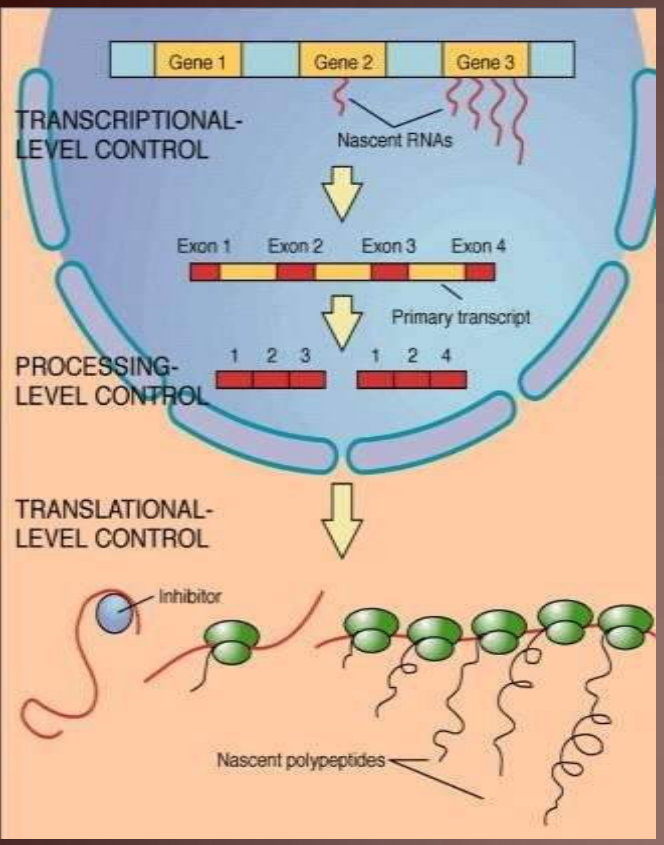


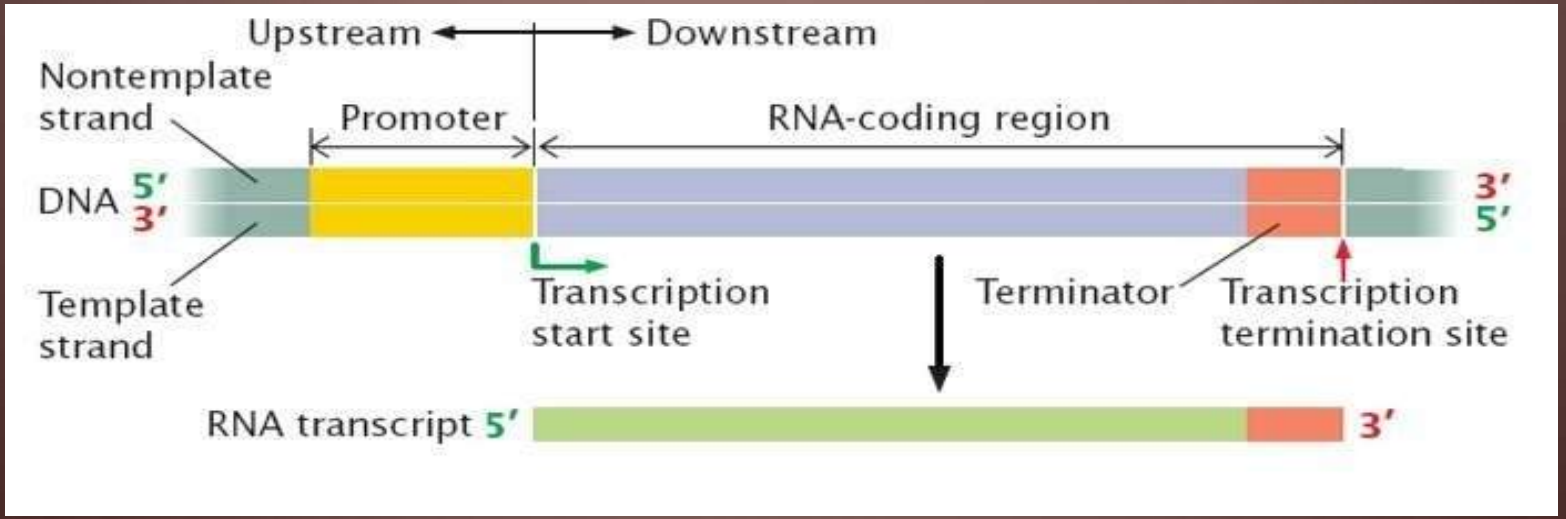
Post transcriptional modifications and Inhibitors of Transcription (DNA Transcription- II)

Contents

- Post Transcriptional Modifications of-
  - Ribosomal RNA
  - Transfer RNA
  - Messenger RNA
- Clinical significance of Splicing
- Biological Significance of Splicing
- Inhibitors of Transcription

Primary Transcript

- A primary transcript is a linear copy of a transcriptional unit-The segment of DNA between specific initiation and termination



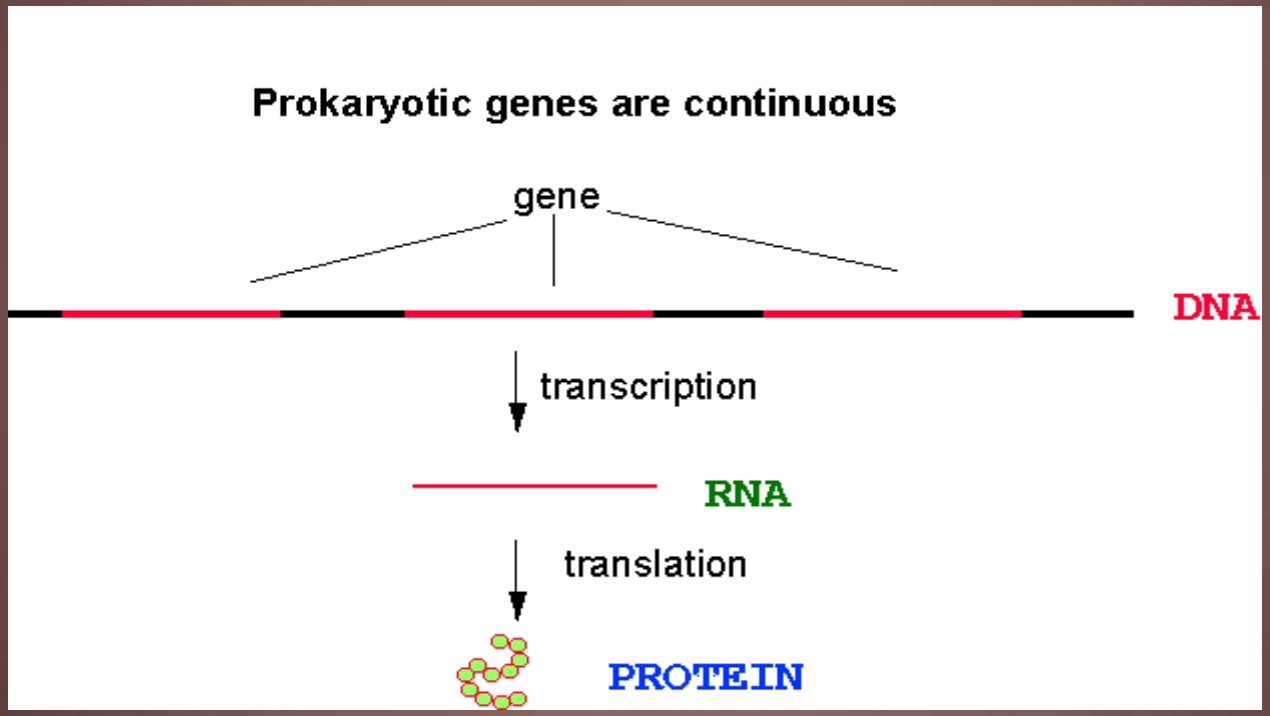
Modifications of prokaryotic primary transcript

- The primary transcript of prokaryotic
- t- RNA s and r -RNA s are post- transcriptionally modified by removing extra nucleotides .
- t RNA s are then further modified to acquire special characters unique to t- RNAs.

Modifications of primary transcript in prokaryotes

- In prokaryotic organisms, the primary transcripts of mRNA-encoding genes begin to serve as translation templates even before their transcription has been completed.
- This is because the site of transcription is not compartmentalized into a nucleus as it is in eukaryotic organisms.
- Consequently, prokaryotic mRNAs are subjected to little processing prior to carrying out their intended function in protein synthesis.

Modifications of primary transcript of mRNA in prokaryotes



- Transcription and translation are coupled in prokaryotic cells.

Post Transcriptional modifications in Eukaryotes

- All eukaryotic RNA primary transcripts undergo extensive processing whether it be as mRNA or as a component of the translation machinery such as rRNA, 5S RNA, or tRNA or RNA processing machinery, snRNAs.
- Processing occurs primarily within the nucleus and includes nucleolytic cleavage to smaller molecules and coupled **nucleolytic and ligation reactions (splicing of exons).**
- The processes of **transcription, RNA processing, and even RNA transport from the nucleus are highly coordinated.**

Processes involved in the post transcriptional modifications

- Some of the processes involved in the post transcriptional modifications of primary transcript of major RNAs are as follows-

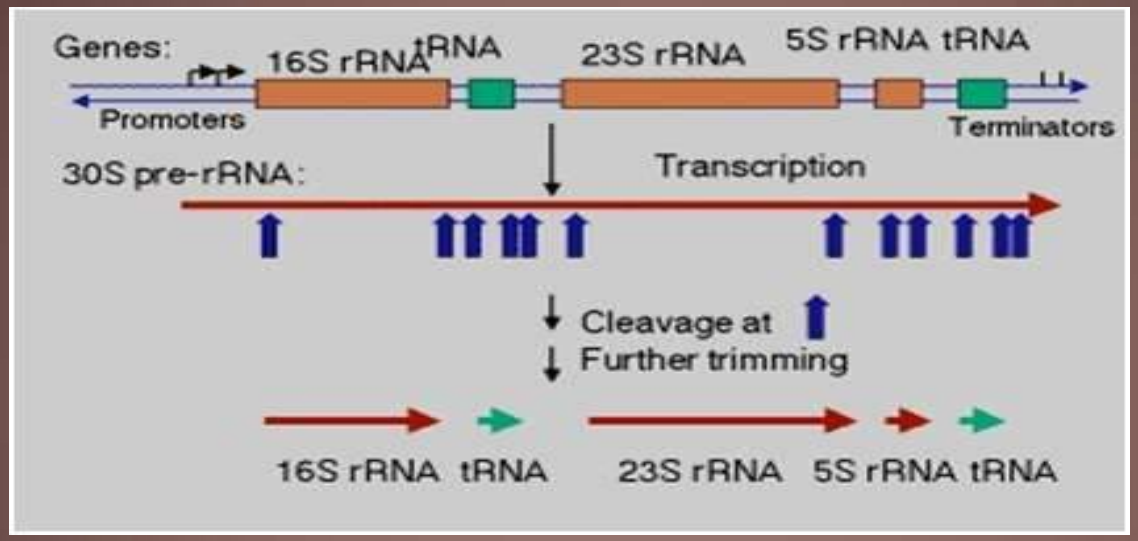
A) Ribosomal RNA

- In mammalian cells, the three rRNA molecules are transcribed as part of a single large precursor molecules called , **Pre ribosomal RNAs**.

Post Transcriptional modifications of Ribosomal RNA(r- RNA)

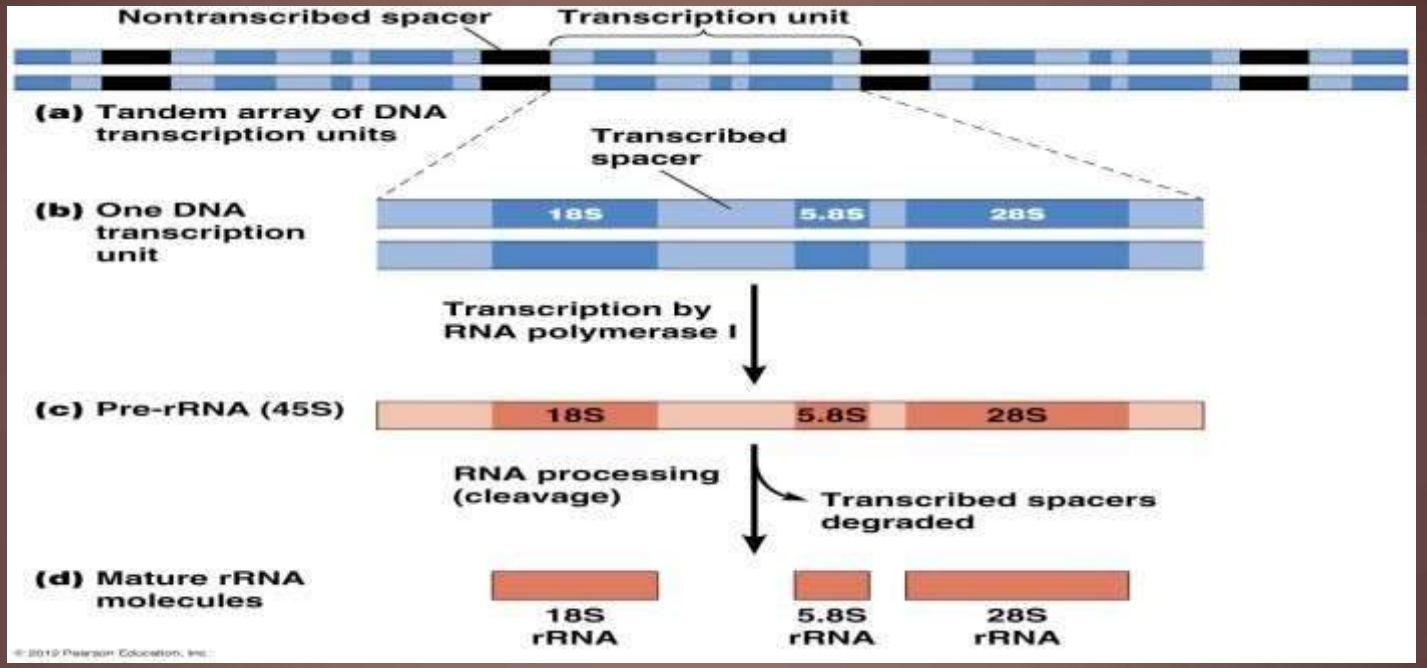
- **The precursor is subsequently processed in the nucleolus** to provide the RNA component for the ribosome subunits found in the cytoplasm.
- The 23S,16S, and 5S ribosomal RNAs of prokaryotes are produced form a single RNA precursor molecule as are the 28S, 18S and 5.8S r RNAs of eukaryotes.
- Eukaryotic 5S rRNA in eukaryotes is synthesized by RNA polymerase III and modified separately.

Post Transcriptional modifications of Ribosomal RNA (r- RNA)



- The 23S,16S, and 5S ribosomal RNAs of prokaryotes are produced form a single RNA precursor molecule
- Cleavage and trimming are the mechanisms involved,
- Similar modifications are observed in the processing of eukaryotic rRNA

Post Transcriptional modifications of Ribosomal RNA(r- RNA)



- 28S, 18S and 5.8S r RNAs are synthesized from a single 45 S precursor in eukaryotes.
- Eukaryotic 5S rRNA is synthesized by RNA polymerase III

Post Transcriptional modifications of Transfer RNA (t- RNA)

- The tRNA molecules serve as adapter molecules for the translation of mRNA into protein sequences.
- Both eukaryotic and prokaryotic transfer RNA s are made from longer precursor molecules that must be modified.
- The basic mechanism involved are as follows-

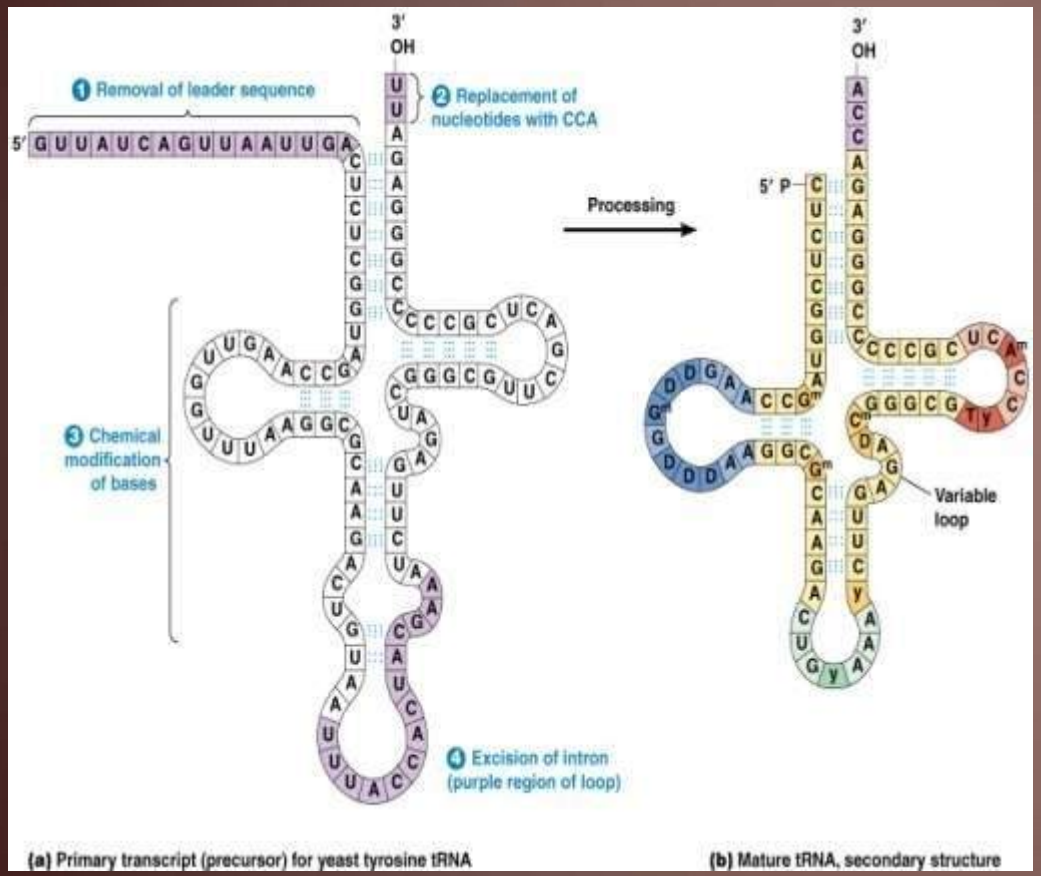
Post Transcriptional modifications of Transfer RNA(t- RNA)

- **Splicing-** An intron must be removed from the anticodon loop
- **Trimming-** The sequences at both the 5' and 3' ends of the molecule are trimmed
- **Base modifications-**The tRNAs contain many modifications of the standard bases A, U, G, and C, including methylation, reduction, deamination, and rearranged glycosidic bonds. Further modification of the tRNA molecules includes nucleotide

Post Transcriptional modifications of Transfer RNA(t- RNA)

- **CCA attachment**
- The attachment of the characteristic CpCpAOH terminal at the 3' end of the molecule by the enzyme nucleotidyl transferase is the most important modification.
- The 3' OH of the A ribose is the point of attachment for the specific amino acid that is to enter into the polymerization reaction of protein synthesis

Post Transcriptional modifications of Transfer RNA(t- RNA)



The extra nucleotides at both 5' and 3' ends of t- RNA are removed, an intron from the anticodon arm is removed, bases are modified and CCA arm is attached to form the mature functional t RNA

Post Transcriptional modifications of pre m-RNA

- In prokaryotic organisms, the primary transcripts of mRNA-encoding genes begin to serve as translation templates even before their transcription has been completed.
- In all eukaryotes the primary transcripts of mRNA-encoding genes undergo extensive processing before they are converted to mature functional forms

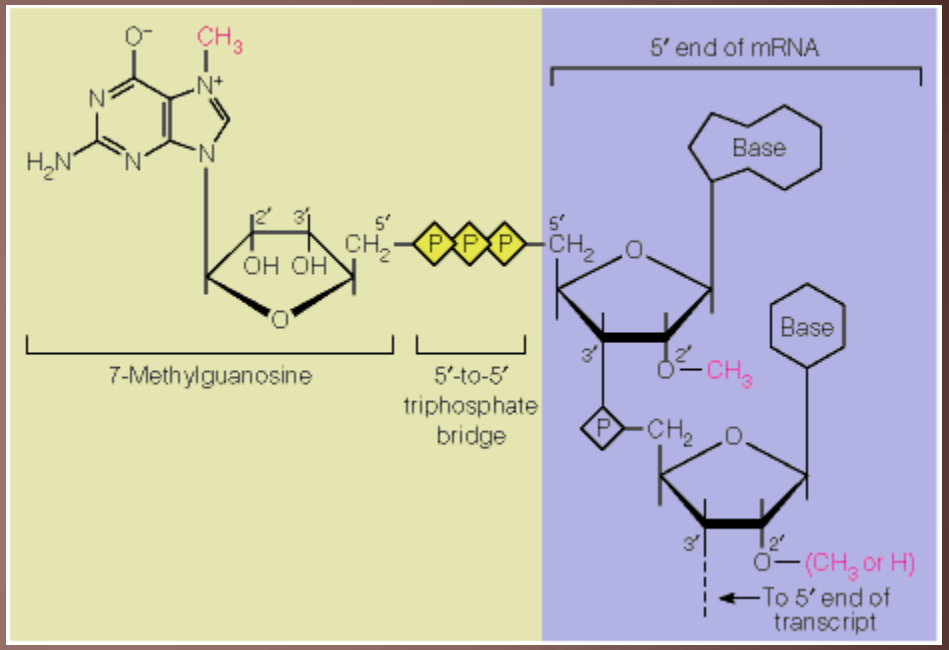
Post Transcriptional modifications of pre m-RNA

a) 5' Capping

- Mammalian mRNA molecules contain a 7-methylguanosine cap structure at their 5' terminal.
- The cap structure is added to the 5' end of the newly transcribed mRNA precursor in the nucleus prior to transport of the mRNA molecule to the cytoplasm.
- The **5' cap** of the RNA transcript is required both for efficient translation initiation and protection of the 5' end of mRNA from attack by 5-'3' exonucleases.
- Eukaryotic m RNAs lacking the cap

Post Transcriptional modifications of pre m-RNA

- The addition of the Guanosine triphosphate (part of the cap is catalyzed by the nuclear enzyme guanylyl transferase.
- Methylation of the terminal guanine occurs in the cytoplasm and is catalyzed by guanine-7-methyl transferase.



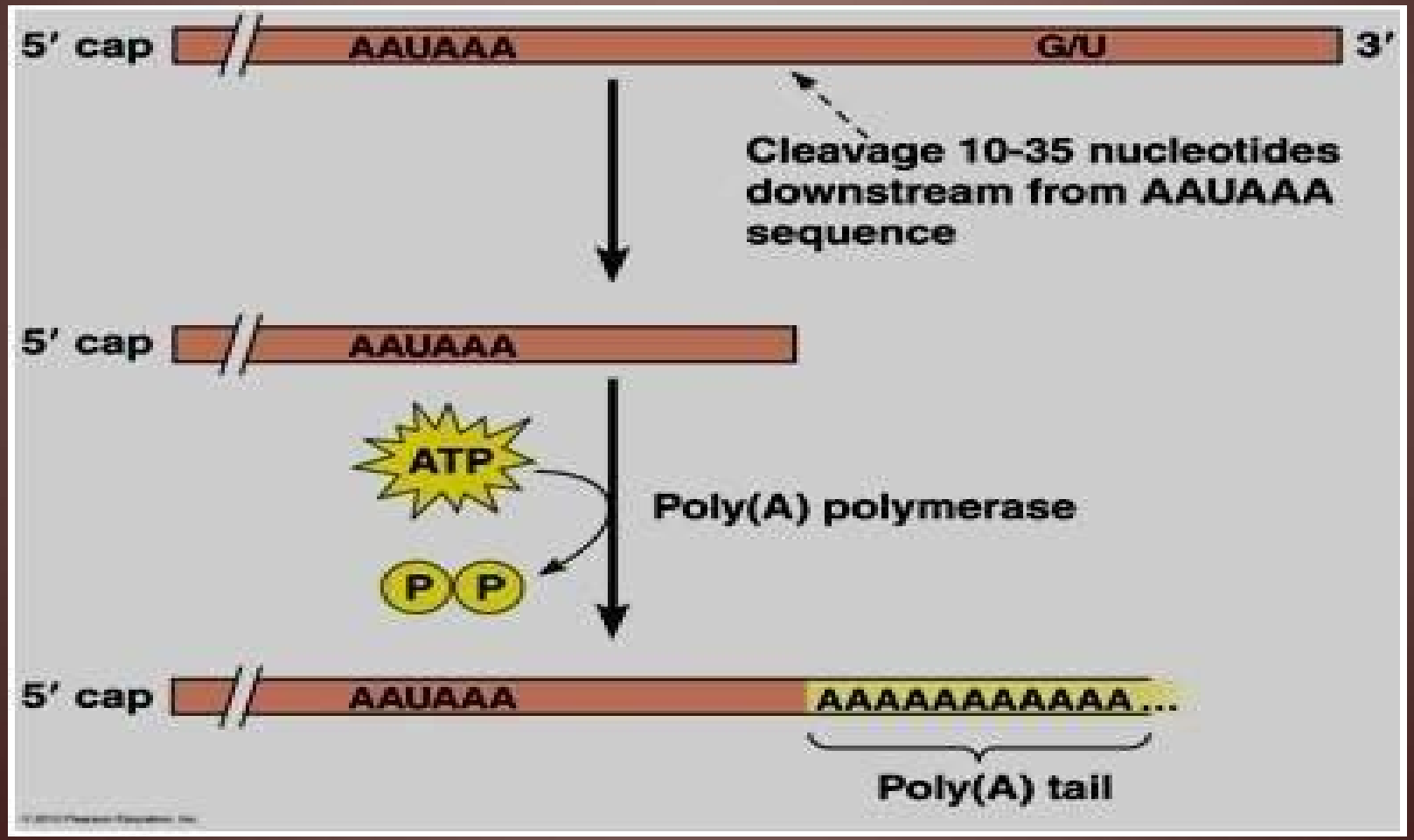
- SAM is the methyl group donor.
- Additional methylation steps may occur.
- The secondary methylations of mRNA molecules, those on the 2'-hydroxy and the N^6 of adenylyl residues, occur after the mRNA molecule has appeared in the cytoplasm.

Post Transcriptional modifications of pre m-RNA

b) Addition of poly A tail

- Poly(A) tails are added to the 3' end of mRNA molecules in a posttranscriptional processing step.
- The mRNA is first cleaved about 20 nucleotides downstream
  from an AAUAA recognition sequence
- Another enzyme, poly(A) polymerase, adds a poly(A) tail which is subsequently extended to as many as 200 A residues.
- The **poly(A) tail** appears to protect the 3' end of mRNA from 3' 5' exonuclease attack.
- Histone and interferon's mRNAs lack poly A tail.
- After the m-RNA enters the cytosol, the poly A tail is gradually shortened.

Post Transcriptional modifications of pre m- RNA

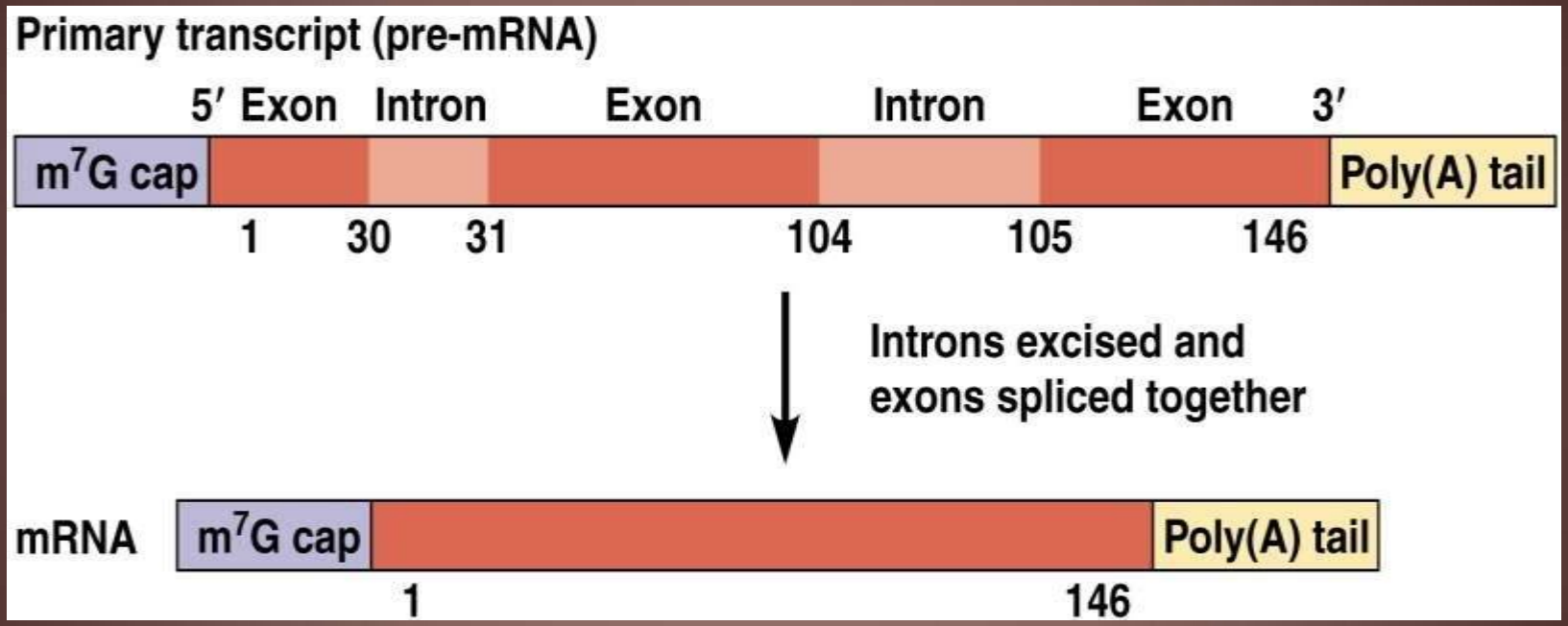


Post Transcriptional modifications of Pre m RNA

Removal of introns (Splicing)

- Introns or intervening sequences are the RNA sequences which do not code for the proteins.
- These introns are removed from the primary transcript in the nucleus, exons (coding sequences) are ligated to form the mRNA molecule, and the mRNA molecule is transported to the cytoplasm.
- The steps of splicing are as follows-

Post Transcriptional modifications of Pre m RNA



- Introns are removed from the primary transcript in the nucleus, exons (coding sequences) are ligated to form the mRNA molecule

Splicing of m-RNA

The steps of splicing are as follows-

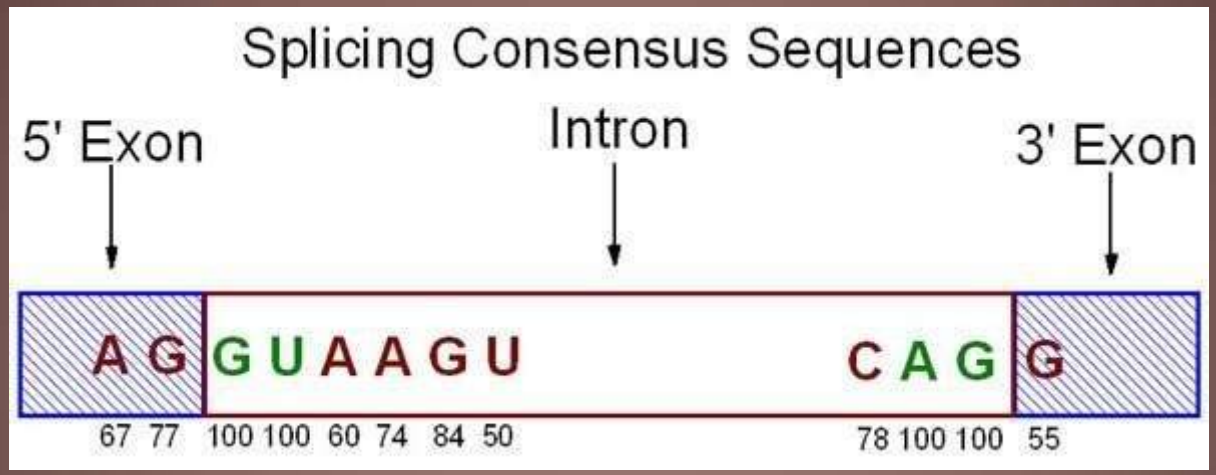
i) <u>Role of small nuclear RNA (sn RNA) and Spliceosome</u>

- The molecular machine that accomplishes the task of splicing is known as the spliceosome. Spliceosomes consist of the primary transcript, five small nuclear RNAs (U1, U2, U4, U5, and U6) and more than 60 proteins.
- Collectively, these form a **small ribonucleoprotein (snRNP) complex,** sometimes called a "**snurp**" **(snRNPs)**

Splicing of m-RNA

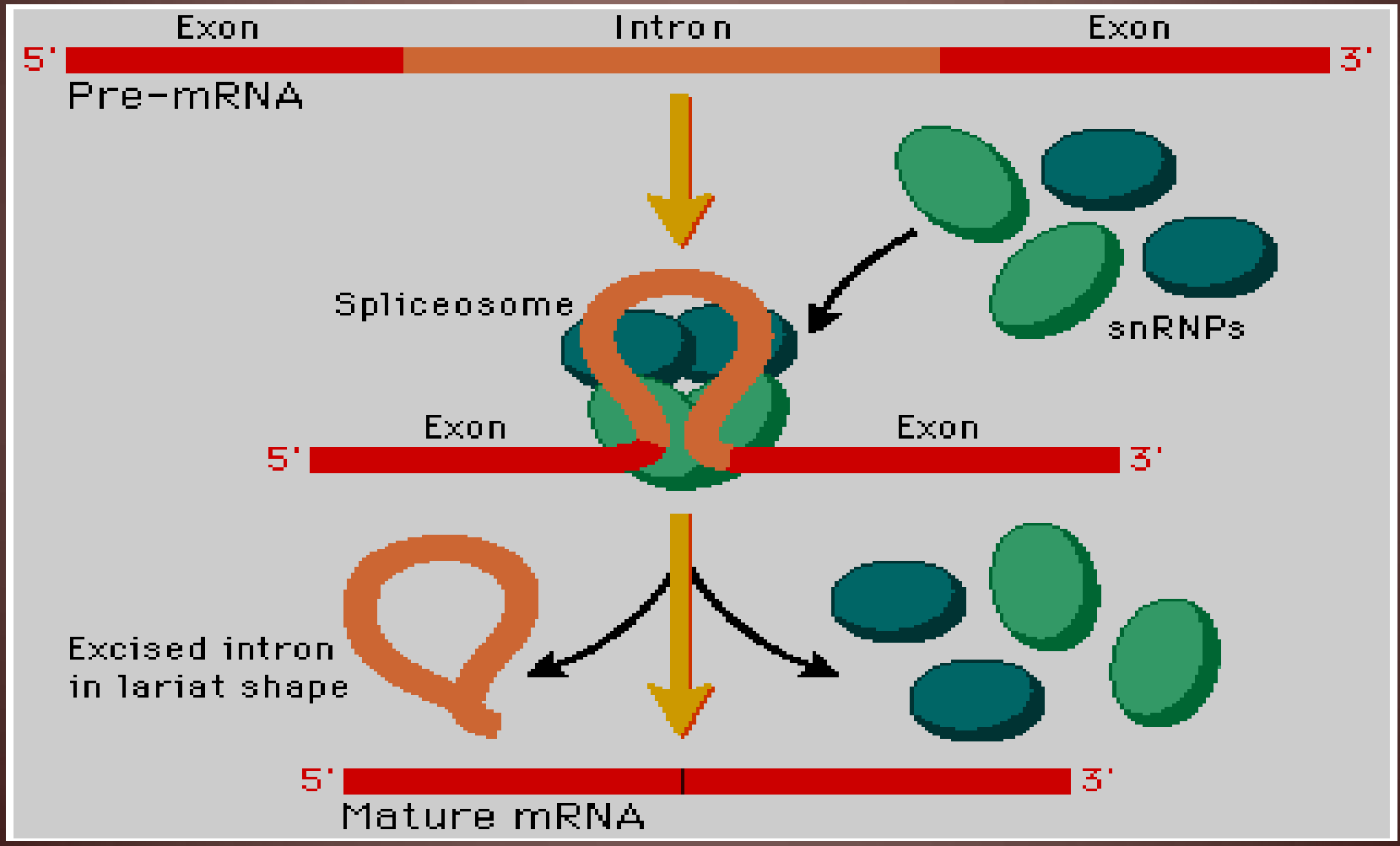
- Snurps are thought to position the RNA segments for the necessary splicing reactions.
- These facilitate the splicing of exon segments by forming base pairs with the consensus sequence at each end of the intron.

Splicing consensus Sequences



There are reasonably conserved sequences at each of the two exon-intron (splice) junctions

Splicing of m-RNA



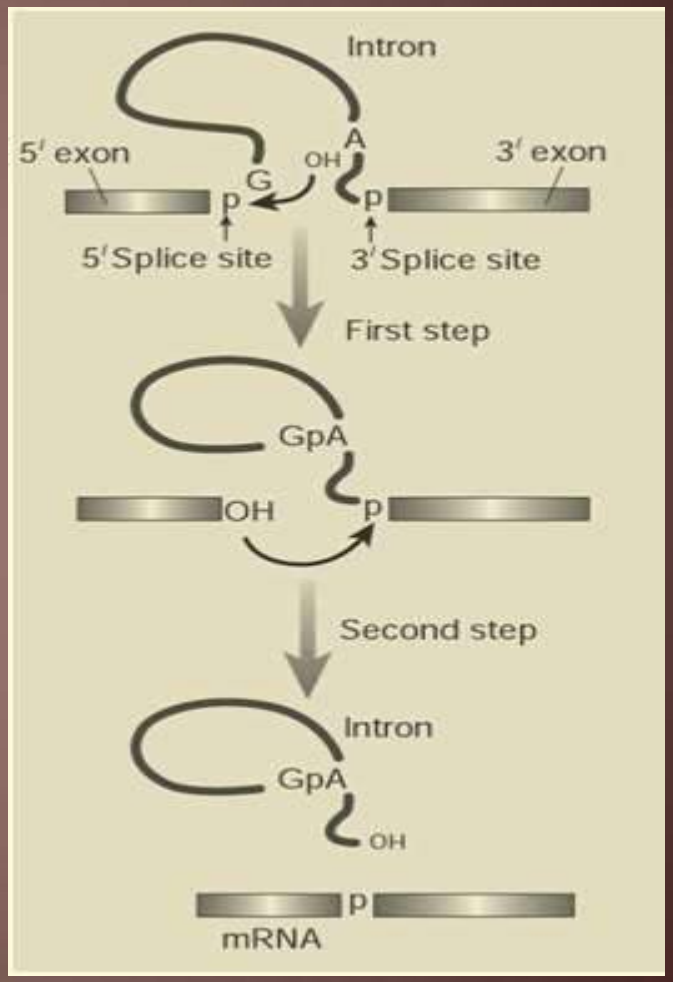
Splicing of m-RNA

ii) Mechanism of excision of introns

- The binding of snRNPs brings the sequences of the neighboring exons in to the correct alignment for splicing.
- The 2'-OH group of an adenosine (A) residue (known as the branch site) in the intron attacks and forms a phosphodiester bond with the phosphate at the 5' end of the intron 1

Splicing of m-RNA

- The newly- feed 3'OH of the upstream exon 1 then forms a phosphodiester bond with the 5'end of the downstream exon 2.
- The excised intron is released as a "lariat" structure, which is degraded
- After removal of all the introns, the mature m RNA molecules leave the nucleus by passing in to the cytosol through pores in to the nuclear membrane.



Clinical significance of Splicing

1) **Antibodies against snRNPs**

In Systemic Lupus Erythematosus (SLE), an auto immune disease, the antibodies are produced against host proteins, including sn RNPs.



Clinical significance of Splicing

2) Mutations at the splice site

- Mutations at the splice site can lead to improper splicing and the production of aberrant proteins .
- For example some cases of Beta thalassemia are as a result of incorrect splicing of beta globin m-RNA due to mutation at the splice site.

Biological significance of Splicing

Alternative Splicing

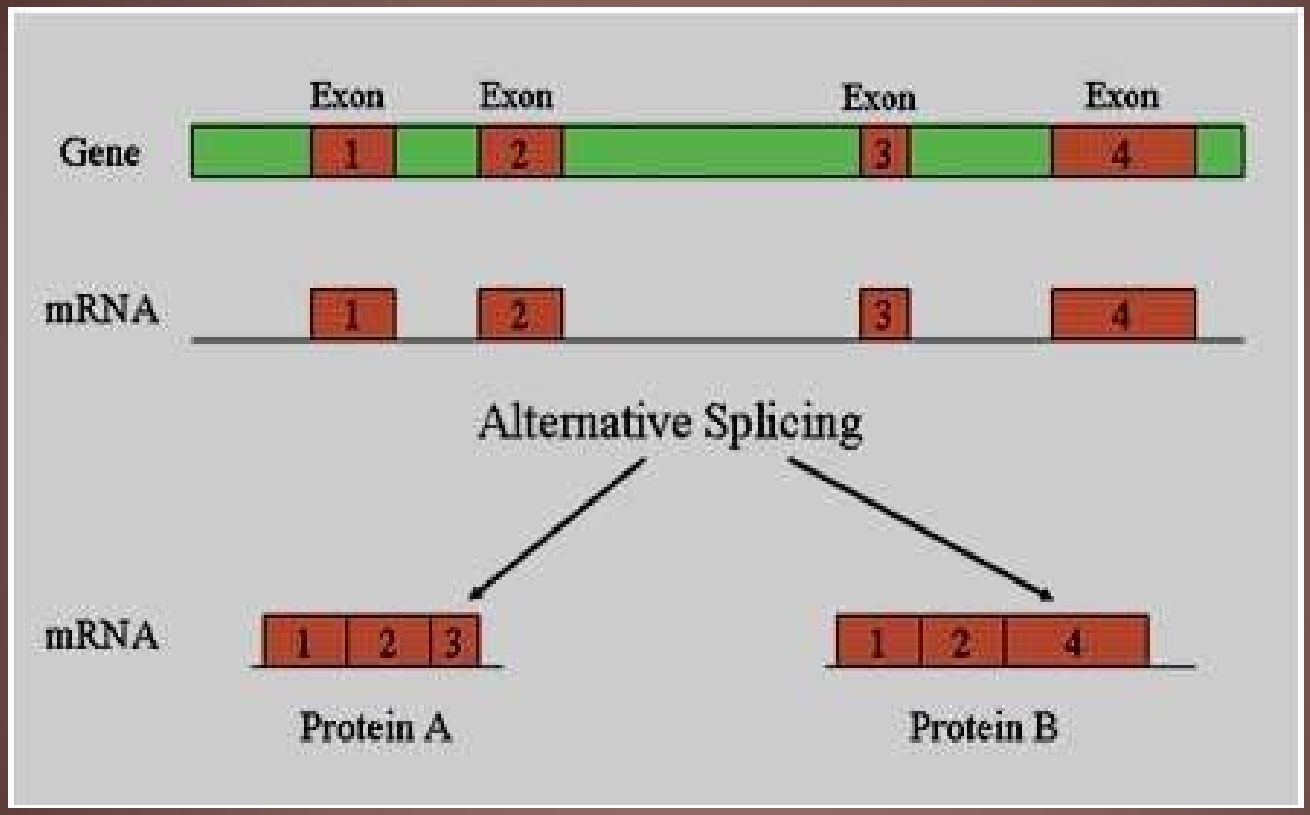
- Alternative patterns of RNA splicing is adapted for the synthesis of tissue-specific proteins.
- The pre-m RNA molecules from some genes can be spliced in two or more alternative ways in different tissues.
- This produces multiple variations of the m RNA and thus diverse set of proteins can be synthesized from a given set of genes

Biological significance of Splicing

Alternative Splicing

- For example- Tissue specific tropomyosins are produced from the same primary transcript by alternative splicing.
- **Alternative splicing and processing** results in the formation of seven unique α -tropomyosin mRNAs in seven different tissues

Biological significance of Splicing

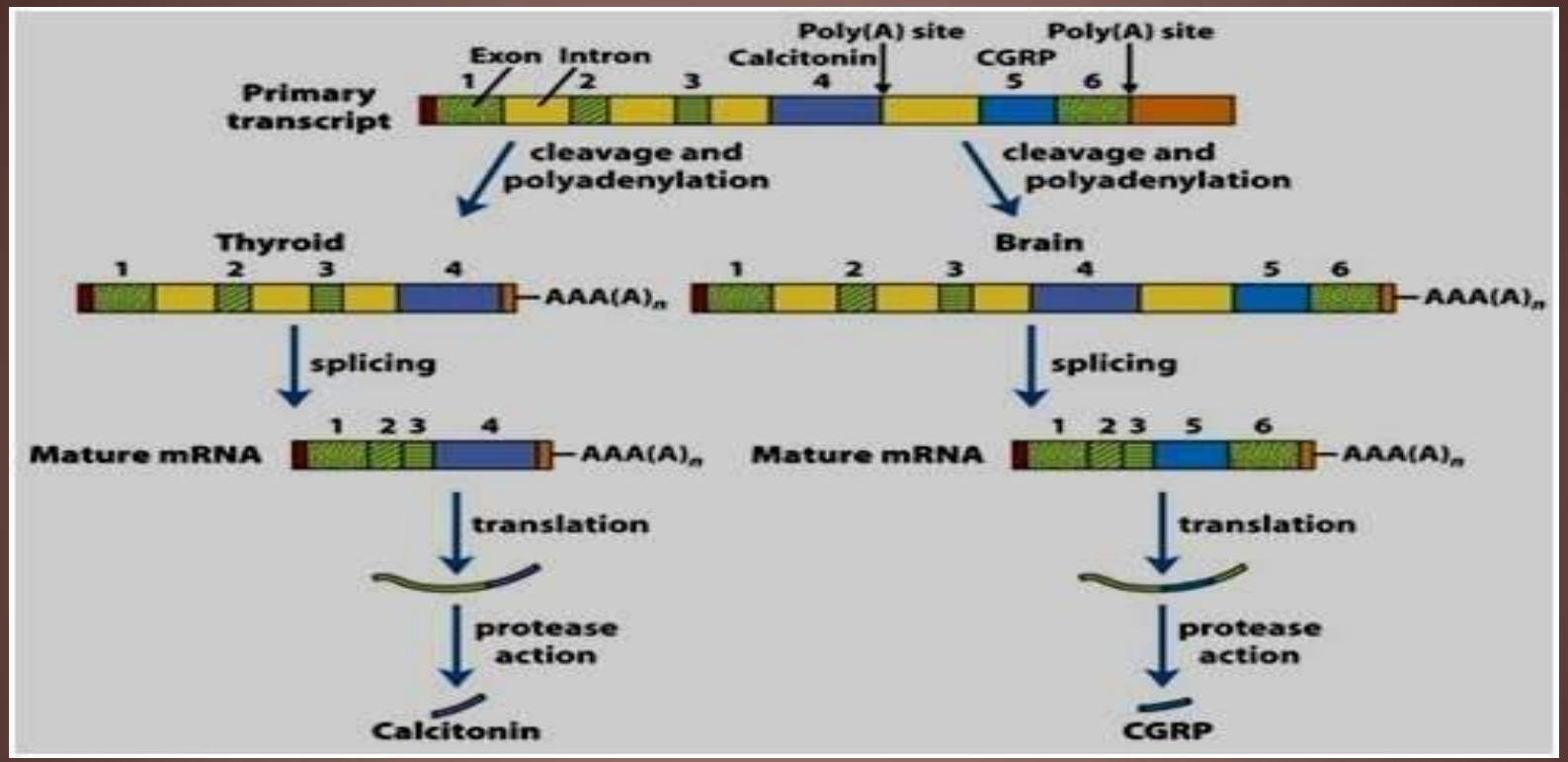


- Tissue specific proteins are producedfrom the same primary transcript by alternative splicing

Biological significance of Splicing

- Similarly, the use of alternative termination- cleavage-polyadenylation sites also results in mRNA variability .
- **Alternative polyadenylation sites** in the immunoglobulin heavy chain primary transcript result in mRNAs that are either 2700 bases long ($_m$) or 2400 bases long ($_s$).
- This results in a different carboxyl terminal region of the encoded proteins such that the $_m$ protein remains attached to the membrane of the B lymphocyte and

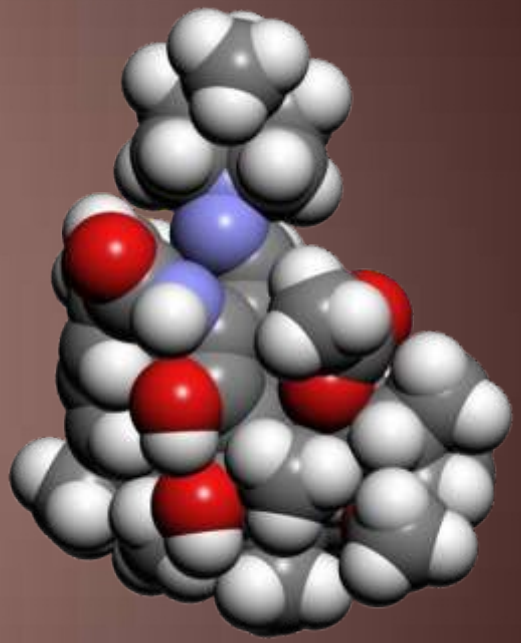
Biological significance of Splicing



- By means of alternative poly A sites variability in mRNA can be produced and thus different proteins can be synthesized from a given set of genes

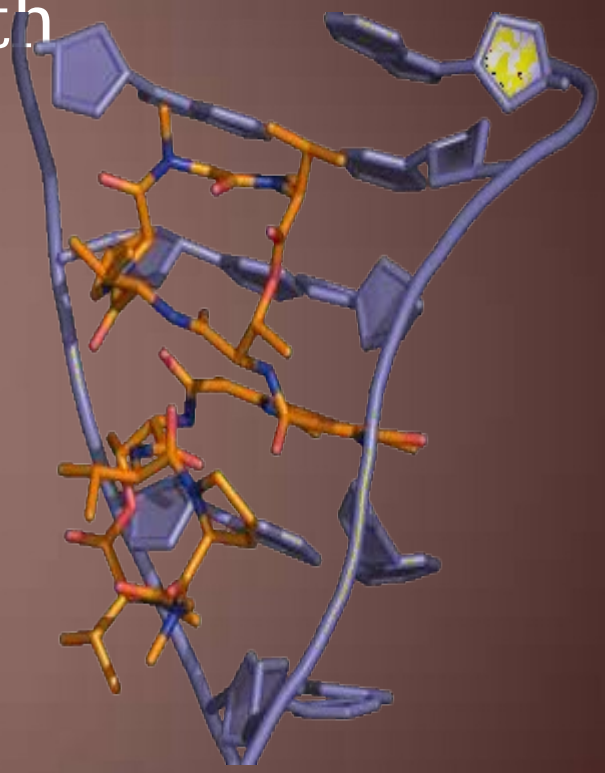
Inhibitors of Transcription

- **Rifampicin**- binds with Beta subunit of prokaryotic RNA polymerase,
- It is an inhibitor of prokaryotic transcription initiation.
- It binds only to bacterial RNA polymerase but not to eukaryotic RNA polymerases.
- Therefore, Rifampicin is a powerful drug for treatment of bacterial infections.
- Used for the treatment of



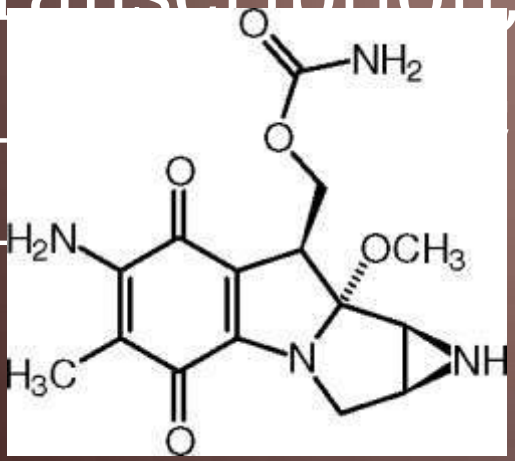
Mechanism of action of Actinomycin D

- Actinomycin D- Intercalates with DNA strands
- Actinomycins inhibit both DNA synthesis and RNA synthesis by blocking chain elongation.
- They interact with G·C base pairs as they require the 2-amino group of guanine for binding.
- Actinomycins are used as anticancer drugs



Mitomycin

- Mitomycin- Intercalates with DNA strands
- blocks transcription,
- u ancer d





Alpha amanitin

- Alpha amanitin is a molecule made from the “death cap” mushroom and is a known potent inhibitor RNA polymerase.
- One single mushroom could very easily lead to a fast death in 10 days.
- The mechanism of action is that alpha amanitin inhibits RNA polymerase –II at both the initiation and elongation states of transcription.



Revision of concepts

- Follow the link

  https://www.khanacademy.org/science/biology/x324d1dcc:metabolism/x324d1dcc:dna-and-rna/v/post-translational-regulation